

Exercise 2.10

Suppose that Gompertz' law applies with

$$\mu_{30} = 0.000130 \quad \text{and} \quad \mu_{50} = 0.000344.$$

Calculate ${}_{10}p_{40}$.

Solution

Under Gompertz' law, the force of mortality has the form

$$\mu_x = Bc^x.$$

Using the two given values,

$$\mu_{30} = Bc^{30} = 0.000130$$

and

$$\mu_{50} = Bc^{50} = 0.000344.$$

Dividing the second equation by the first gives

$$\begin{aligned} \frac{Bc^{50}}{Bc^{30}} &= \frac{0.000344}{0.000130}, \\ c^{20} &= \frac{0.000344}{0.000130}. \end{aligned}$$

Therefore,

$$c = \left(\frac{0.000344}{0.000130} \right)^{1/20} = 1.049858.$$

Then

$$B = \frac{0.000130}{c^{30}} = 3.020095 \times 10^{-5}.$$

Now,

$${}_{10}p_{40} = S_{40}(10) = \exp \left\{ - \int_{40}^{50} \mu_r \, dr \right\}.$$

Since $\mu_r = Bc^r$,

$$\begin{aligned} {}_{10}p_{40} &= \exp \left\{ - \int_{40}^{50} Bc^r \, dr \right\} \\ &= \exp \left\{ -B \left[\frac{c^r}{\log c} \right]_{40}^{50} \right\} \\ &= \exp \left\{ -\frac{B}{\log c} (c^{50} - c^{40}) \right\}. \end{aligned}$$

Substituting $B = 3.020095 \times 10^{-5}$ and $c = 1.049858$,

$$\begin{aligned} {}_{10}p_{40} &= \exp \left\{ -\frac{3.020095 \times 10^{-5}}{\log(1.049858)} (1.049858^{50} - 1.049858^{40}) \right\} \\ &= 0.997279876. \end{aligned}$$

Thus,

$$\boxed{{}_{10}p_{40} \approx 0.997280}.$$